DO NOW — RECALL

Lesson 065 · Computational Methods: Recognising Solva

Name: _____ Class: _____ Date: _____

1. What is a precondition?

2. What is abstraction?

3. How does a queue differ from a stack?

4. Trace binary search for 11 in the sorted, zero-indexed array [2,4,6,8,10,12,14]. Start low=0, high=6. Use mid=(low+high) DIV 2 and exclude the checked midpoint when updating bounds. State the checked indexes and the final result.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Condition required for an algorithm to work correctly.

2. Model answer

Selecting relevant details and omitting irrelevant ones.

3. Model answer

FIFO versus LIFO.

4. Model answer

3, 5, 4; not found. Values are 8, 12, 10. The final bounds are low=5, high=4, so the search stops.

By the end, you can…

• A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

• Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

• Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

Vocabulary — know these terms

problem recognition — Identify the underlying task type.

constraint — Requirement a solution must obey.

objective function — Quantity to minimise or maximise.

computable specification — Precise task that a program can implement/check.

NEW LEARNING

Read, annotate and explain

Identify

A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

Model

Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

Evaluate

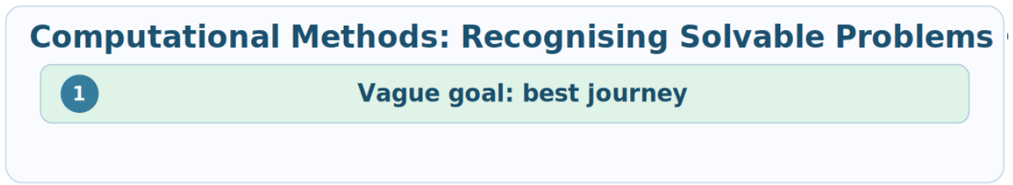
Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

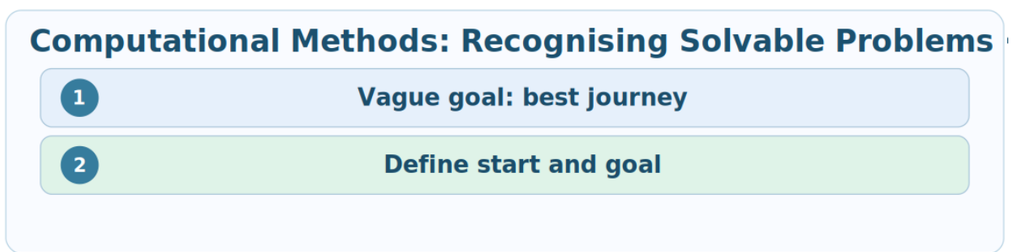
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



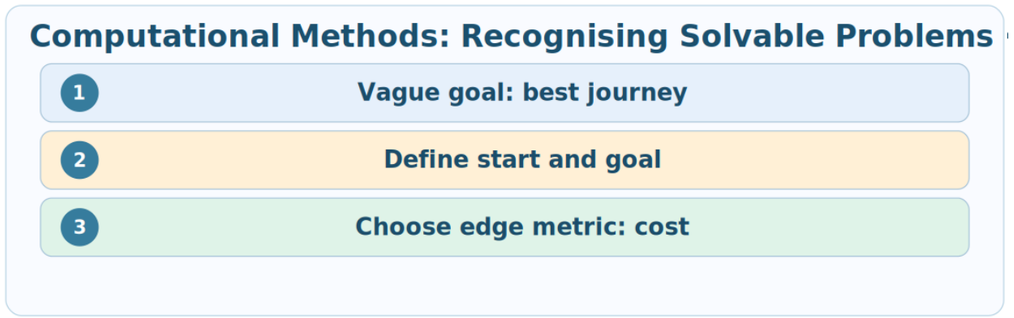
Computational Methods: Recognising Solvable Problems · STEP 1: Vague goal: best journey

New Learning



Computational Methods: Recognising Solvable Problems · STEP 2: Vague goal: best journey; Define start and goal

New Learning



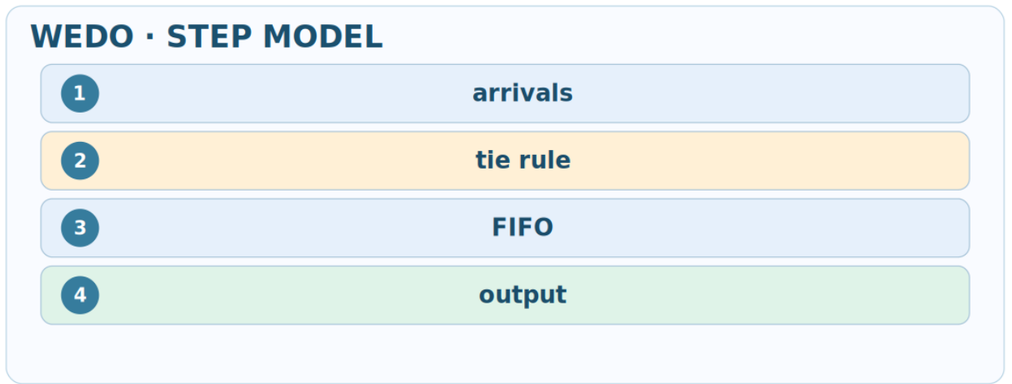
Computational Methods: Recognising Solvable Problems · STEP 3: Vague goal: best journey; Define start and goal; Choose edge metric: cost

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Turn 'pick the fairest lunch queue order' into a computable FIFO rule.

An ordering rule makes fairness measurable and repeatable within the chosen model.



WEDO · STEP MODEL: arrivals; tie rule; FIFO; output

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

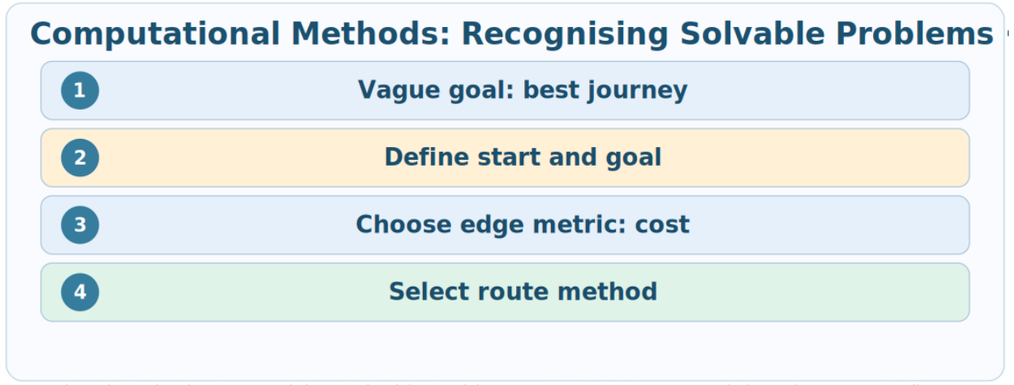
Guided 1 — CHECK

Explain the decision
An ordering rule makes fairness measurable and repeatable within the chosen model.

New Learning · original lesson diagrams

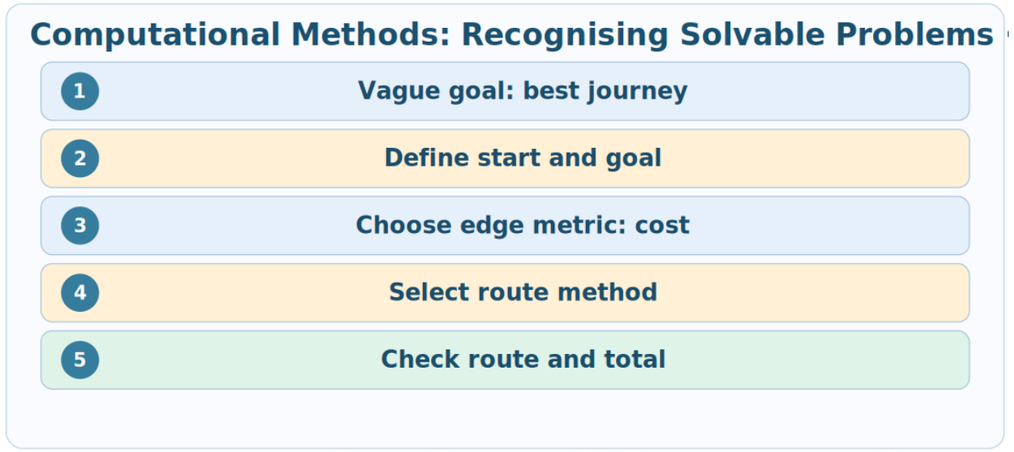
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Computational Methods: Recognising Solvable Problems · STEP 4: Vague goal: best journey; Define start and goal; Choose edge metric: cost; Select route method

New Learning



Computational Methods: Recognising Solvable Problems · STEP 5: Vague goal: best journey; Define start and goal; Choose edge metric: cost; Select route method; Check route and total

I DO — WORKED EXAMPLE

Follow the worked method

Turn 'choose the best bus route' into a precise computational problem.

Explain the decision

Cost is defined and connections must be valid; output stops and total fare.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Every vague human objective has exactly one obvious numerical optimisation target.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate whether a computer can choose a 'good teacher' simply from exam results, explaining models, constraints and limits. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Turn 'choose the best bus route' into a precise computational problem.

Cost is defined and connections must be valid; output stops and total fare.



IDO · STEP MODEL: start; dest; fares; minimum cost

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Turn 'pick the fairest lunch queue order' into a computable FIFO rule.
Source code / input:
Input ordered arrival timestamps and ticket IDs; output same order, oldest first; handle ties by sequence number.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Define computational problem input, process, output for cheapest route on a network. [6]

Question 2

A pupil asks to 'sort books by best'. Give two precise orderings, explain why different and select one for finding a title. [6]